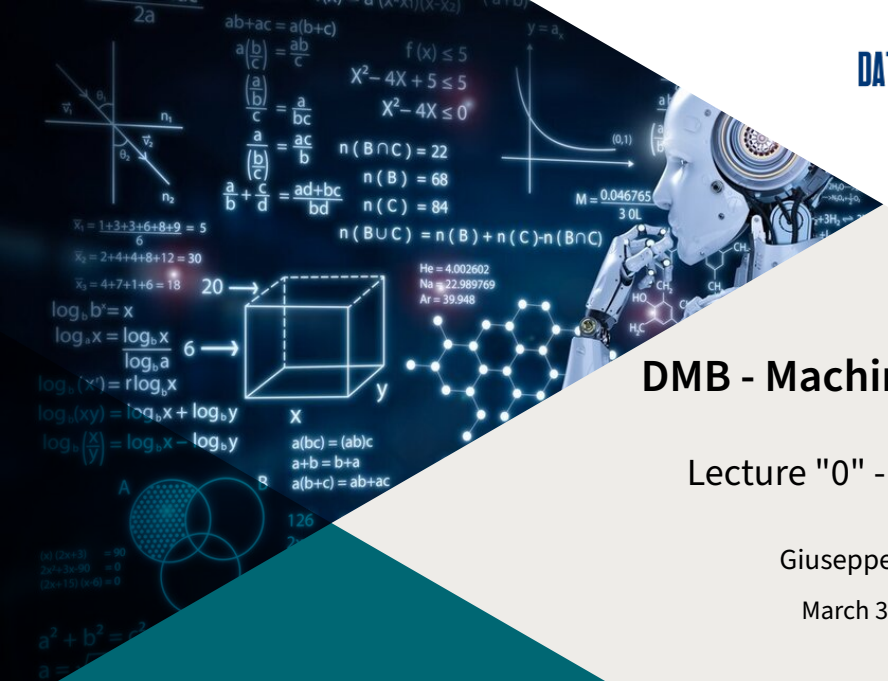

DMB - Machine Learning

Lecture "0" - The course

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Prerequisites and some info on the course

Why?

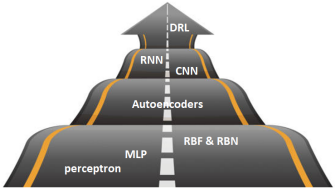


Figure 1: The road to become a data scientist (©G.Longo).

If you are reading this slide, it means that you want to become proficient in machine learning for Data Science. This is not an easy journey for two reasons:

- AI tools are better than most humans at programming and at solving complex formal problems. Hence, you are obsolete, unless... you realize that
- data science is both complex and a highly inter-disciplinary topic which requires a good knowledge of mathematics, statistics and programming. Only in this case you will be able to beat AI.

We assume that you already know the foundations of statistics, and the foundations of linear algebra. We also assume that you have some familiarity with Python programming and know how to exploit some basic libraries such as SciKit and Panda.

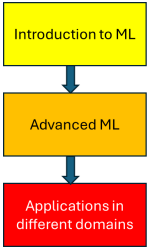
WARNING

This course is structured in **three parts**:

- Introduction to machine learning
- Advanced Machine Learning
- Applications of Machine learning to different domains.

This course relies on several resources:

- These slides are just a support to Some reference textbooks (see below)
- A full set of Jupyter Notebooks which contains detailed text and python examples and codes
- Examples of full scale projects
- Links to relevante references, on-line material, etc



The Exam - I

To take the exam you must have passed the following exams

- Statistical Data Analysis (both modules)
- Hardware and Software

The Exam for the entire course (module A and B) is single and covers ALL topics covered in the two modules.

The Exam consists of written multichoice tests and an oral exam where a **team of 2 persons** (find a companion no individual exams will be allowed) must present a project.

The **projects will be assigned by the professors**. If you think that a topic might be of special interest to you, you can propose it, but the final decision is on the professors.

During the courses, there will be 2 multiple choice tests for each module that will be evaluated for the final mark.

The Exam - II

attendance at the lectures will be signed at the beginning of the lecture, in the first quarter of the hour after the official starting time of the lecture. **Checks will be made for fake signatures**, and the culprits will be expelled from the course (no possibility to take the exam during this academic year).

If you attend 80% of lectures and do all the tests (*) (scoring better than 60%) for each module during the course, then you can directly access the oral exam. Otherwise, you must do a written multichoice test on the overall program of both modules. If you score better than 60% on this test, you will be admitted to the oral exam.

For **understating the methodology**, we mean that you must understand and know the **assumptions, applicability, the limitations and the MATHEMATICS** behind the methodologies you will be dealing with.

The Exam - III

- The oral exam will consist of presenting and discussing the methodological aspects of a project work on a case study. Since that of data scientist is collaborative work, it will be **necessary to be in two people** to develop the project work. One week before the exam session, you are asked (MUST) to send to both professors a short presentation in .ppt or .pdf of the project, the code you have written and the technical report with scripts and methodological explanation
- **Important Notice**: all your materials will be checked for authenticity and percentage of AI contribution (using Turnitin, Google Gemini, and other professional tools). Hence, if you use snippets of code produced by AI in the code you MUST comment on the corresponding section of the code, specifying which was the source, what was your request to the LLM of your choice, and justify why you made such a request. The criteria we shall adopt for the grading are in the following table.
- The presentation (in ppt or PDF) must describe all steps of the project: problem definition and understanding; exploratory data analysis; criteria followed in selecting a specific method/model; description of the experiments; discussion of the results.

The Exam - IV

Criterion Exam Grade	Knowledge of the contents	Rigor in applying concepts and methods	Quality of the Reporting and the Script	Clarity of the Exposure
< 18 Not sufficient	*fragmented and superficial*	*fundamental errors*	*insufficient*	*insufficient*
18 - 20	*sufficient but general concepts*	*uncertainties in the application of concepts and methods*	*simplified reporting and sufficient script*	*elementary exposure*
21, 22, 23, 24	*appropriate but not in-depth knowledge of contents*	*ability to apply fundamental concepts*	*ability of reporting in a simple way and good script*	*clear and correct presentation*
25, 26, 27	*precise and complete knowledge of contents*	*ability to apply broad and complete concepts*	*ability of reporting in an articulated way and very good script*	*safe and correct exposure*
28, 29, 30, 30 cum laude	*very broad, complete, and in-depth knowledge of contents*	*well-established ability to apply contents and excellent ability to analyze, synthesize and interdisciplinary connections*	*ability to present broad, complete and in-depth contents in an articulated way and excellent script*	*mastery of exposure*

Figure 2: GRADING SYSTEM.

Materials

This course aims at being selfconsistent, but studying such a complex topic relies heavily on your personal efforts in deepening your knowledge. This course draws heavily from some seminal books which you should definitely buy and keep in your library. They will prove crucial for your professional future and will be a lifelong resource. These reference books are:

- C.M. Bishop & H. Bishop, Deep Learning, Foundations and concepts, 2024. Springer. **This is the main book of the course**.
- Deep Learning, Y. Goodfellow, Y. Bengio and A. Courville, MIT press, Cambridge (2016). Hereafter shortened as [GBC];
- Deep Learning with Python, 2. Ed, F. Chollet, Manning Publ. C. (2021) [CHO]
- Charu A. Aggarvall, Neural Networks and Deep Learning, Springer, 2018 [AG2]

Other useful references (available on line) will be linked to the slides, while others (not available on line) will be listed where needed at the end of the units.

Useful sites

- Some sites are an useful source of information, tips, updates on the rapidly evolving field of data sciences and machine learning. Remember, however, that most blogs are written by students like you, or by non professionals who use these posts in the hope to gain some recognition. **MOST OF THE TIMES THEY CONTAIN wrong or incomplete results**.
- **Useful sites**: Data Science Central; KDnuggets; AISummer;
- Register to the **COLAB** platform operated by Google. If you need to acquire some understanding of the platform's basic features, watch this video;
- Register to **GITHUB** and learn how to fully exploit their huge number of services.

Software to be installed

In order to practice with the methods we shall encounter during the course you need to have the following softwares installed on your PC and to register to a few sites.

- **Anaconda Navigator**. Is an interface which makes easy to install Python and R libraries, create environments, etc. To familiarize with it, it can be useful to watch this video.
- Once you have installed Anaconda, from its pop-up window install **Jupyter Notebook**. To acquire some familiarity watch this video.
- For more complex task it might prove usefule to acquire some familiarity also with Visual Studio Code (highly recommended).

Some mathematical notations

Main mathematical notations

- Vectors are denoted by lower case bold roman letters such as $\mathbf{x}$. All vectors are assumed to be column vectors unless otherwise stated.
- $\mathbf{1}$ denotes a column vector in which all elements have the value 1.
- Matrices are denoted by uppercase bold roman letters, such as $\mathbf{M}$.
- $det\mathbf{M}$ is the determinant of the matrix $\mathbf{M}$.
- A superscript $^\top$ denotes the transpose of a matrix or vector, so that $\mathbf{x}^\top$ will be a row vector.
- The notation $(w_1, \ldots, w_M)$ denotes a row vector of M elements and the corresponding column vector is $\mathbf{w} = (w_1, \ldots, w_M)^\top$.
- The $M \times M$ identity matrix (also known as the unit matrix) is denoted as $\mathbf{I}_M$. It is defined as $I_{ij} = \delta_{ij}$, and will be abbreviated to $\mathbf{I}$ if there is no ambiguity.
- The concatenation of two vectors $\mathbf{a}$ and $\mathbf{b}$ is denoted with $\mathbf{a} \oplus \mathbf{b} = a_1, \ldots, a_N, b_1, \ldots, b_M)$

- The notation $x \sim p(x)$ signies that x is sampled from the distribution $p(x)$. Where there is ambiguity, we use subscripts as in $p_x(.)$ to denote which density is referred to.
- The expectation of a function $f(x, y)$ with respect to a random variable x is denoted by $\mathbb{E}_x\left[f(x, y)\right]$.
- If the distribution of x is conditioned on another variable z , then the corresponding conditional expectation will be written $\mathbb{E}_x\left[f(x)|z\right]$.
- The variance of $f(x)$ is denoted $var\left[f(x)\right]$.
- For vector variables, the covariance is written $cov\left[x, y\right]$.
- On a graph, the set of neighbours of node i is denoted $\mathcal{N}(i)$, which should not be confused with the Gaussian or normal distribution $\mathcal{N}(x|\mu, \sigma)$.
- $\{\ \}$ are used to identify a set.
- The notation $g(x) = \mathcal{O}(f(x))$ denotes that $|f(x)/g(x)|$ is bounded as $x \to \infty$. For instance, if $g(x) = 3x + 2$, then $g(x) = \mathcal{O}(x^2)$.
- The notation $\lfloor x \rfloor$ denotes the "floor" of x, i.e., the largest integer that is $\leq x$.

- $\frac{d}{dx}$ - derivative with respect to x
- $\frac{\partial}{\partial x}$ - partial derivative with respect to x
- $\nabla_{\mathbf{x}} y$ - gradient of y with respect to $\mathbf{x}$
- $\nabla_X y$ - matrix derivatives of y with respect to X
- $\nabla_{\mathbf{X}} y$ - tensor containing derivatives of y with respect to $\mathbf{X}$
- $\frac{\partial f}{\partial \mathbf{x}}$ - Jacobian matrix $J \in \mathbb{R}^{mxn}$ of $f : \mathbb{R}^n \to \mathbb{R}^m$
- $\nabla^2_{\mathbf{x}} f(\mathbf{x})$ or $\mathbf{H}(f)(\mathbf{x})$ - the Hessian matrix of f at input point $\mathbf{x}$
- $\int f(\mathbf{x}) d\mathbf{x}$ - definite integral over the entire domain of $\mathbf{x}$
- $\int_{\mathbb{S}} f(\mathbf{x}) d\mathbf{x}$ - definite integral with respect to over the set S